WORKSHEET — UNIT I

Python Basics

Course: Data Science with Python **Unit:** I (10 Hrs)

Name: _____ **Roll No:** _____ **Date:** _____

Topics covered: Data types • Loops & comprehensions • Functions • Modules • Classes • Exceptions

Section 1 — Key Concepts (Quick Revision)

Term	Simple Meaning	Example
Variable	A name that stores a value.	`x = 10`
Dynamic typing	Python finds the type by itself. You do not declare it.	`x = 5 then x = "hi"`
List	Ordered items. Can be changed.	`[1, 2, 3]`
Tuple	Ordered items. Cannot be changed.	`(1, 2, 3)`
Dictionary	Stores data as key : value pairs.	`{"name": "Ravi"}`
Loop	Repeats a block of code.	`for i in range(3):`
List comprehension	A short way to build a new list.	`[x*x for x in range(3)]`
Function	A reusable block of code with a name.	`def add(a, b):`
Module	One .py file that you can import.	`import math`
Class & Object	Class = blueprint. Object = item made from it.	`class Dog: ...`
Operator overloading	Make + work on your own objects using __add__.	`def __add__(self, other):`
Exception	An error while the program runs. Handle it with try/except.	`try: ... except: ...`

Section 2 — Multiple-Choice Questions

Choose the correct option and circle the letter.

1. **Which of these data types cannot be changed after creation (immutable)?**
   - **A.** List **B.** Dictionary
   - **C.** Tuple **D.** Set
2. **What does type(5/2) give in Python 3?**
   - **A.** int **B.** float
   - **C.** str **D.** complex
3. **Which keyword is used to define a function?**
   - **A.** func **B.** function
   - **C.** def **D.** lambda
4. **What is the output of [x*x for x in range(3)] ?**
   - **A.** [0, 1, 2] **B.** [0, 1, 4]
   - **C.** [1, 4, 9] **D.** [0, 2, 4]

5. **Which function shows output on the screen?**
   A. print() **B.** input()
   C. show() **D.** echo()

6. **Which line imports only the sqrt function from the math module?**
   A. import math.sqrt **B.** from math import sqrt
   C. import sqrt from math **D.** include math.sqrt

7. **Inside a class method, what does self mean?**
   A. The parent class **B.** The class itself
   C. A loop keyword **D.** The current object

8. **Which special method is used to overload the + operator?**
   A. __plus__ **B.** __sum__
   C. __add__ **D.** __concat__

9. **Which block runs whether or not an error occurs?**
   A. raise **B.** try
   C. except **D.** finally

10. **What is the output of: for i in range(1, 4): print(i, end=" ") ?**
    A. 1 2 3 **B.** 0 1 2
    C. 1 2 3 4 **D.** 0 1 2 3

Section 3 — Fill in the Blanks

Word bank: list raise module __init__ except

1. A _____ is an ordered collection that can be changed. A tuple cannot be changed.
2. The _____ keyword is used to raise an error manually.
3. A _____ is a single .py file that can be imported into another program.
4. The _____ method runs automatically when a new object is created.
5. The _____ block catches and handles errors from the try block.

Section 4 — True or False

1. Python is a statically typed language. **(True / False)**
2. A dictionary stores data as key–value pairs. **(True / False)**
3. A tuple can be changed after it is created. **(True / False)**
4. A list comprehension can build a new list from an existing one. **(True / False)**
5. The finally block runs whether or not an error occurs. **(True / False)**

Section 5 — Match the Following

Write the correct letter from Column B next to each item in Column A.

No.	Column A	Ans.	Column B
1	List		a. Ordered collection that cannot be changed
2	Tuple		b. Reusable block of code
3	Dictionary		c. Single .py file that can be imported
4	Function		d. Collection of key–value pairs
5	Module		e. Error that can be caught using try/except
6	Exception		f. Ordered collection that can be changed

Section 6 — Short-Answer Questions

Answer each question in 1–2 sentences.

1. **What is dynamic typing? Give one example.**

2. **Write one difference between a list and a tuple.**

3. **What is a default argument in a function? Give an example.**

4. **What is the difference between a module and a package?**

5. **Why do we use try/except? Give one example.**

Section 7 — Practical Programs

Write and run each program in Python. Use the hint if you get stuck.

Practical 1: Find the even numbers

1. Make a list of 10 numbers (1 to 10).
2. Use a list comprehension to keep only the even numbers.
3. Print the new list.

Hint:

```
numbers = [1, 2, 3, 4, 5, 6, 7, 8, 9, 10]
even = [ ___ for x in numbers if ___ ]
print(even)

# Tip: x % 2 == 0 means x is even
```

Expected output: *[2, 4, 6, 8, 10]*

Practical 2: Rectangle class

1. Create a class Rectangle with length and width (use __init__).
2. Add a method area() that returns length × width.
3. Create two rectangles and print their areas.

Hint:

```
class Rectangle:
    def __init__(self, length, width):
        self.length = ___
        self.width = ___

    def area(self):
        return ___

r1 = Rectangle(4, 5)
print(r1.area())
```

Expected output: *The area of each rectangle (for example 20).*

Practical 3: Safe division

1. Write a function divide(a, b).
2. Use try/except to catch ZeroDivisionError.
3. If b is 0, return a message such as "Cannot divide by zero".

Hint:

```
def divide(a, b):
    try:
        return a / b
    except ___:
        return "Cannot divide by zero"

print(divide(10, 2))
print(divide(10, 0))
```

Expected output: *5.0 and Cannot divide by zero*

Instructor Answer Key

Section 2 — MCQ

Q	Answer	Why
1	C	A tuple cannot be changed after it is created.
2	B	The / operator always gives a float in Python 3.
3	C	def starts a function definition.
4	B	0, 1, 2 squared gives [0, 1, 4].
5	A	print() displays output on the screen.
6	B	from module import name imports one item.
7	D	self refers to the object calling the method.
8	C	Python calls __add__ when + is used on objects.
9	D	finally always runs, with or without an error.
10	A	range(1, 4) gives 1, 2, 3 (the end value is not included).

Section 3 — Fill in the Blanks

1. list 2. raise 3. module 4. __init__ 5. except

Section 4 — True or False

1. False 2. True 3. False 4. True 5. True

Section 5 — Match the Following

1 → f 2 → a 3 → d 4 → b 5 → c 6 → e

Section 6 — Short Answers (Key Points)

- Python decides the type of a variable from its value while the program runs. Example: x = 5, then x = "hi" is allowed.
- A list can be changed and uses [ ]. A tuple cannot be changed and uses ( ).
- A default argument is a value used when the caller gives none. Example: def greet(name="Guest").
- A module is one .py file. A package is a folder of modules (with an __init__.py file).
- It stops the program from crashing. Example: catch FileNotFoundError when a file is missing.

Section 7 — Practical Programs (Sample Solutions)

Practical 1

```
numbers = [1, 2, 3, 4, 5, 6, 7, 8, 9, 10]
even = [x for x in numbers if x % 2 == 0]
print(even)    # [2, 4, 6, 8, 10]
```

Practical 2

```
class Rectangle:
    def __init__(self, length, width):
        self.length = length
        self.width = width

    def area(self):
        return self.length * self.width

r1 = Rectangle(4, 5)
r2 = Rectangle(3, 7)
print(r1.area())    # 20
print(r2.area())    # 21
```

Practical 3

```
def divide(a, b):
    try:
        return a / b
    except ZeroDivisionError:
        return "Cannot divide by zero"

print(divide(10, 2))    # 5.0
print(divide(10, 0))    # Cannot divide by zero
```